

# Reproducing LoRA on RoBERTa-base

CS 4782 Final Project Summary

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## 1. Goal

LoRA freezes pretrained weights and learns a low-rank update  $\Delta W = BA$  on selected matrices, aiming to match full fine-tuning with far fewer trainable parameters. We reproduce Hu et al.’s RoBERTa-base GLUE result and test whether the low intrinsic-rank claim holds on MRPC. We then ask a further question: can LoRA absorb an attention-level architectural change, Exclusive Self-Attention (XSA), when inserted only at fine-tuning time?

## 2. Setup

We target the RoBERTa<sub>base</sub> + LoRA result on GLUE MRPC from Hu et al. (2021), then rerun the rank ablation  $r \in \{1, 2, 4, 8\}$  on MRPC. As a secondary task, we also run SST-2 at  $r=8$ . This slice tests both the parameter-efficiency claim—LoRA gives most of full fine-tuning’s benefit with few trainable weights—and the intrinsic-rank claim—larger  $r$  should not matter much once the update has enough rank.

We implement LoRA directly by wrapping each attention query and value projection in RoBERTa-base with a trainable low-rank update, freezing the original linear weights and training only LoRA parameters plus the classifier head. We train with AdamW, batch size 32, max length 128, seed 42, and linear warmup/decay. LoRA uses learning rate  $5 \times 10^{-4}$ ; full fine-tuning uses  $2 \times 10^{-5}$ . MRPC runs for 3 epochs, while SST-2 runs for 2000 steps due to wall-clock budget. We also include a frozen head-only baseline to verify that LoRA’s gains come from low-rank adaptation rather than the classifier alone.

## 3. Reproduction results

On MRPC, full fine-tuning reaches 88.2%, LoRA  $r=8$  reaches 86.3%, LoRA  $r=1$  reaches 86.0%, and the head-only baseline reaches 68.4%. Although our LoRA result is below the paper’s reported 87.5%, the 18-point gap over head-only confirms that the low-rank update, not just the new classifier, is doing the adaptation.

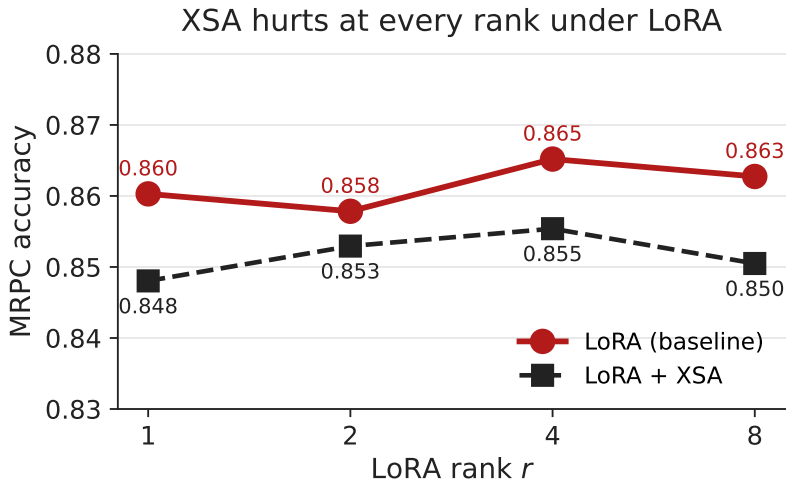
| $r$              | 1     | 2     | 4     | 8     |
|------------------|-------|-------|-------|-------|
| Trainable params | 629K  | 666K  | 739K  | 887K  |
| MRPC accuracy    | 0.860 | 0.858 | 0.865 | 0.863 |

The spread across ranks is only 0.7 points, within the noise expected on the 408-example MRPC validation set. This reproduces the qualitative finding that increasing  $r$  does not unlock much additional performance, suggesting that the useful fine-tuning update has low intrinsic rank. On SST-2, LoRA  $r=8$  reaches 93.1% in roughly one epoch, compared with the paper’s 95.1% after a full sweep. We therefore match the paper’s main trend, though not its exact absolute accuracy, likely because our runs are single-seed and lightly tuned.

## 4. Additional exploration: LoRA under Exclusive Self-Attention

We also tested whether LoRA can absorb an attention-level architectural change at fine-tuning time. Exclusive Self-Attention (XSA) projects each head’s attention output away from its own value vector,  $Y \leftarrow Y - (Y \cdot V_n)V_n$ , with  $V_n = V/\|V\|$ , to reduce self-similarity bias. Because XSA was proposed for from-scratch training, we asked whether a pretrained RoBERTa-base model could tolerate the change when only LoRA and the classifier head are trainable.

|               | MRPC $r=1$ | $r=2$ | $r=4$ | $r=8$ | SST-2 $r=8$ |
|---------------|------------|-------|-------|-------|-------------|
| Baseline LoRA | .860       | .858  | .865  | .863  | .931        |
| LoRA + XSA    | .848       | .853  | .855  | .850  | .922        |
| $\Delta$      | -1.2       | -0.5  | -1.0  | -1.3  | -0.9        |



XSA consistently hurts by about one accuracy point in our single-seed runs. The runs remain stable, so the issue is not optimization collapse; rather, XSA changes the representation distribution in every attention layer after pretraining. This suggests that XSA is not a drop-in fine-tuning-time modification for standard pretrained RoBERTa under a small LoRA budget. A natural follow-up is full fine-tuning with XSA to test whether the gap is due to adaptation capacity rather than the XSA constraint itself.

## 5. Takeaways

Our reproduction supports the qualitative LoRA claim: a tiny low-rank update to  $W_q$  and  $W_v$  gives most of the benefit of full fine-tuning, while rank has little effect on MRPC once  $r \geq 1$ . The main mismatch with the paper is absolute accuracy, likely due to single-seed fixed-hyperparameter runs. The XSA experiment is a useful negative result: LoRA trains stably under the modified attention rule, but does not fully compensate for the layerwise distribution shift introduced after pretraining.

References: Hu et al. (2021), *LoRA: Low-Rank Adaptation of Large Language Models*; Liu et al. (2019), *RoBERTa*; Wang et al. (2019), *GLUE*; Aghajanyan et al. (2020), *Intrinsic Dimensionality Explains the Effectiveness of Language Model Fine-Tuning*.